ACCENTRA: A MULTILINGUAL SPEECH-TO-SPEECH TRANSLATION AND STRUCTURED POLICE COMPLAINT ASSISTANCE FRAMEWORK

PHASE I REPORT

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BONAFIDE CERTIFICATE

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"This project addresses the following Sustainable Development Goal(s): Goal 10 – Reduced Inequalities."

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ABSTRACT

In this project, we present the Accentra system which consists of accent adaptation to a multilingual model for speech-to-text translation with police complaint assistance in the languages Tamil, Hindi, Telugu, and Malayalam in India. Accentra addresses communication difficulties encountered during conveying information for police in the cases where there is no shared language between the people and the officials to whom information needs to be conveyed. In the baseline automatic speech recognition system, the IndicConformer (600 Million Parameters, Hybrid CTC-RNNT) pretrained on a large corpus is fine-tuned using a handpicked subset of AI4Bharat IndicVoices dataset covering six regions. The data include recordings of spontaneous speech varying in duration between 2 and 20 seconds per individual per language, capped by number of speaker samples at about 20 per person. Overall, 5,075 recordings (about 10.96 hours) were divided into disjoint per-speaker training (3,951 samples), validation (347) samples, and test sets (777). The baseline is the pretrained model mentioned above, that is compared against an adapted model with regard to performance in an unseen test set. The WER and CER metrics are used in order to evaluate the model. For translation of the recognized native text into English, we utilize the IndicTrans2 Indic-to-English distillation model. Information Extraction GLiNER (a transformer-based model), is applied to the English police complaint draft for obtaining the fields of complainant, location, date, time, persons involved, and property as well as producing the summary of complaint and draft in English language. Fields not populated in the input are left blank. All the parts of the pipeline were designed for CPU-only computation, without external APIs of generative commercial AI.

Keywords: ASR, Accent Adaptation, IndicConformer, IndicVoices, IndicTrans2, GLiNER, speech translation, information extraction, police complaint assistance.

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TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	**iii**
	LIST OF TABLES	**vii**
	LIST OF FIGURES	**viii**
	LIST OF ABBREVIATIONS	**ix**
1.	**INTRODUCTION**	**1**
	1.1 GENERAL	1
	1.2 OBJECTIVE	2
	1.3 EXISTING SYSTEM	3
	1.4 PROPOSED SYSTEM	5
2.	**LITERATURE SURVEY**	**7**
	2.1 LITERATURE REVIEW	7
	2.2 COMPARISON AND DISCUSSION	11
	2.3 CONCLUSION	12
3.	**SYSTEM DESIGN**	**14**
	3.1 SYSTEM ARCHITECTURE	14
	3.2 SYSTEM REQUIREMENTS	16
	3.2.1 SOFTWARE REQUIREMENTS	16
	3.2.2 HARDWARE REQUIREMENTS	17
	3.2.3 DATASET REQUIREMENTS	18
	3.2.4 DEPLOYMENT AND SCALING	19
	3.3 SYSTEM DESIGN	20
	3.3.1 ACTIVITY DIAGRAM	20
	3.3.2 DATA FLOW DIAGRAM	21
4.	**PROJECT DESCRIPTION**	**23**
	4.1 METHODOLOGIES	23
	4.2 MODULES	23
	4.2.1 DATA MANAGER PACKAGE	23
	4.2.2 CORE PACKAGE	24
	4.2.3 TRANSLATION PACKAGE	25

4.2.4 COMPLAINT ASSISTANT PACKAGE 26
4.2.5 UI PACKAGE 26
5. IMPLEMENTATION AND RESULT DISCUSSION 28
5.1 IMPLEMENTATION RESULTS 28
5.1.1 IMPLEMENTATION OF DATA PRE-PROCESSING 28
5.1.2 IMPLEMENTATION OF ASR ADAPTATION 30
5.1.3 IMPLEMENTATION OF TRANSLATION AND COMPLAINT EXTRACTION 31
5.2 DISCUSSION 35
6. CONCLUSION AND FUTURE WORK 37
6.1 CONCLUSION 37
6.2 FUTURE WORK 37
REFERENCES 39

LIST OF TABLES

TABLE NO.	TABLE NAME	PAGE NO.
2.1	Summary of the Literature Survey	11
3.1	Software Requirements of the Proposed System	15
3.2	Hardware Requirements of the Proposed System	16
3.3	Dataset Selection Criteria	17
3.4	Regional Locations Represented in the Dataset	17
3.5	Dataset After Speaker Balancing	17
3.6	Speaker-Disjoint Data Split	18
3.7	Language-wise Speaker-Disjoint Split	18
4.1	ASR Fine-Tuning Configuration	23
4.2	Structured Complaint Fields	25
5.1	WER and CER of the Baseline and Adapted ASR on the Test Split	28
5.2	Extracted Information for the Example Complaint	29

LIST OF FIGURES

FIGURE NO.	FIGURE NAME	PAGE NO.
1.1	Overview of the Proposed System Architecture	5
3.1	Regional Dataset Preparation Pipeline	13
3.2	ASR Fine-Tuning and Evaluation Design	14
3.3	General Speech Translation Pipeline	14
3.4	Police Complaint Assistant Pipeline	15
3.5	Activity Diagram of the Application	19
3.6	Level-0 Data Flow Diagram	20
3.7	Level-1 Data Flow Diagram	20
5.1	Clips per Language Before and After Speaker Balancing	26
5.2	Duration of Balanced Data per Language	27
5.3	Language-wise Speaker-Disjoint Split	27
5.4	Overall Proportion of Training, Validation and Testing Data	28

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
ASR	Automatic Speech Recognition
BERT	Bidirectional Encoder Representations from Transformers
BPCC	Bharat Parallel Corpus Collection
CER	Character Error Rate
CPU	Central Processing Unit
CTC	Connectionist Temporal Classification
DFD	Data Flow Diagram
GLiNER	Generalist Model for Named Entity Recognition
GPU	Graphics Processing Unit
LLM	Large Language Model
NER	Named Entity Recognition
NLP	Natural Language Processing
NMT	Neural Machine Translation
RNNT	Recurrent Neural Network Transducer
SDG	Sustainable Development Goal
UI	User Interface
WER	Word Error Rate

CHAPTER 1
INTRODUCTION

1.1 GENERAL

India boasts remarkable linguistic diversity and is considered one of the most linguistically diverse countries in the world. People often move from one place to another in pursuit of education, work, and travel into areas where their mother tongue is neither commonly used nor known. Sometimes it becomes sufficient to have some understanding of the local language while conducting ordinary conversation. However, there are situations in which complete and detailed knowledge of language is essential.

A good area here would be when a citizen is in dealings with law-enforcement agents. For instance, when traveling to another state from one's place of origin and has an experience to report regarding theft or any form of property damage. But one might not know the language of the law-enforcement agency or even English, therefore necessitating translation services or the need to communicate with what little knowledge of the language available. In such instances, important information could be left out or reported in a wrong way.

The Automatic Speech Recognition (ASR) technique and Neural Machine Translation (NMT) provide a solution for overcoming these barriers in communication. The first one converts the speech into the text, while the latter technique converts the text written in a certain language into the target language. Combining both of them together will allow a person to speak in their mother tongue and generate understandable content for the other person.

Nevertheless, there are some difficulties faced while dealing with Indian languages. There is a considerable difference between people speaking the same language owing to the fact that:

- Region and dialect in geography.
- Accent and speaking speed.
- expressions that have an informal feel,

- Age, gender, and the other speaker features,
- code switching naturally from one's own language into English and
- Environmental factors such as background sounds and microphone quality.

The pre-trained generic speech recognition model is likely to have been trained using a broad coverage of the distribution of speech, which may not give similar results to regional speech distribution. With that in mind, the model developed through the Accentra framework has multilingual speech recognition with region speech adaptation. The method involves the utilization of a pre-trained Indic Conformer, which serves as the base for automatic speech recognition and which is fine-tuned using a regional subset of the AI4Bharat IndicVoices dataset, which contains regional languages of Tamil, Hindi, Telugu, and Malayalam. The adapted ASR model is the speech recognition model in Accentra, while the IndicTrans2 is used for translating the speech into English.

Accentra besides general translation of speech includes a Police Complaint Assistant, which extracts valuable information from the translation of the police complaint using GLiNER, an Information Extraction (IE) Model based on transformers. The extracted information is presented as a structured police complaint in English language format. Thus, the project includes:

1. multi-language automated speech recognition.
2. Accent & Regional Dialects Through Optimization of Automatic Speech Recognition Technology
3. Indo-Aryan language into English Translating:
4. Complaint knowledge in specialized domain.
5. Transformational information extraction.
6. Formalized complaint production, and
7. Online users' interactions.

1.2 OBJECTIVE

The first goal is to build an accent-adaptive multilingual speech-to-text processing system able to transcribe the speech in Indian regional languages into the English language and utilize the processed text in real-world application of police

complaint assistance. It will be necessary for the developed system to recognize speech in Tamil, Hindi, Telugu, and Malayalam languages using multilingual ASR technology and fine-tune the initial pre-trained ASR model so that it adapts to the accentual specifics of regional dialects. Additionally, it is planned to create a well-balanced speech corpus based on data provided by AI4Bharat Indic Voices by selecting audio samples from various locations across the country for each of the four languages and minimizing excessive representation of any of the speakers via speaker balancing procedure. In order to reliably evaluate the ASR process the developed system will employ separate training, validation, and testing datasets for different speakers. For comparative analysis of the pretrained ASR model and fine-tuned ASR models, the Word Error Rate (WER) and Character Error Rate (CER) will be used. Finally, the texts produced by the model in Tamil, Hindi, Telugu, and Malayalam will be translated into English; it will be important for the system to work with naturally spoken and incorrect speech and support code-switching. The second objective of the project is related to the building of the Police Complaint Assistance for situations where language mismatch occurs between citizens and police officers. The assistant should be able to extract relevant complaint information from the text, structurize the complaint fields from unstructured speech and produce a meaningful English summary as well as the text draft of complaint report. Moreover, it will be implemented a CPU-oriented inference pipeline in order to avoid long-term usage of GPU hardware and avoid dependency on the generative AI tools API.

1.3 EXISTING SYSTEM

The currently known multilingual communication techniques include the use of human interpreters, translations, speech translation techniques, and speech recognition systems. They have been described below with limitations inherent to each technique respectively.

1.3.1 Human Translation. In this case, one needs to have a bilingual person who understands both languages to help in translating the dialogue between the citizen and the policeman. Sometimes, it may be impossible to find such a translator.

1.3.2 Text-based translation. The text-based translation system involves the ability of the user to type in information through a translation tool. The issues faced in this case

include the following: a person may encounter difficulty in typing his native language or describing an incident.

1.3.3 General Speech translation. Modern-day speech translation can convert the spoken information from one language into another. However, the process produces the translation results in texts form only. A police report contains lots of important information including the complainant's name, address, details of the incident, date and time, parties involved, and items that were stolen.

1.3.4 Pre-trained Automatic Speech Recognition (ASR) models. The pre-trained ASR models are capable of high-quality and reliable general speech recognition. However, because they are trained on the wider distribution of speech, they may experience problems with some regional phonetics characteristics. Therefore, it may be insufficient to use a pre-trained ASR model out-of-the-box for solving regionality issues with Indian languages.

1.3.5 Complaints structure manual. In cases when English language comes from the speech translator system, it has to be manually extracted and then converted into a complaint form by the police. This is a time-consuming procedure and makes them vulnerable to errors.

The limitations of the existing methods as identified in this research project are listed below:

- language communication barriers due to differences between the languages,
- reliance on the human translator availability,
- difficulties in understanding dialects and accentuated language,
- generic automatic speech recognizer systems that do not make any regional adjustments for the specific language,
- orally communicated inputs that bring trouble to transcription,
- generalized translation without taking into account the context of the complaints,
- absence of automation for pulling complaints,

- manual identification of the complaint details, and dependency of the specific artificial intelligence solution on third-party software programming interfaces and significant computational resources.

Such limitations led to the development of the system as described below

1.4 PROPOSED SYSTEM

Accentra offers an integrated system of multilingual speech processing. The main difference from traditional systems lies in the usage of a fine-tuned multilingual ASR model, instead of the initial pretrained one. There are four major stages of research in this paper.

Phase 1- Preparation of a region-specific corpus of speech. The spoken corpora for Tamil, Hindi, Telugu, and Malayalam were gathered using the AI4Bharat IndicVoices repository. Six districts from each language were chosen, followed by filtering of audio data in line with our project needs, balance of speakers, and creation of datasets using a speaker disjoint division.

Phase 2 - fine tuning and adaptation of ASR model. Pretrained multilingual model of Indic Conformer 600M (ai4bharat/indic-conformer-600m-multilingual) acts as the pretraining checkpoints and then proceeds through fine tuning on regional training data. The baselines are set according to pretraining model, whereas the proposed adapted ASR models are obtained after the fine tuning. WER and CER are calculated on identical test data.

Phase 3 – General translation. User speaks one of four languages; the customized IndicConformer system produces an English-language transcript, and then the IndicTrans2 tool translates it into English.

Phase 4 – Police Complaint Assistant. The English text of the complaint is inserted into GLiNER, who then extracts the information regarding complaints. This information is structured in fields in order to produce the summarized form of complaints in addition to an English format of the complaints.

Figure 1.1 gives an overview of the proposed system. The upper half shows the offline dataset preparation and adaptation stages, and the lower half shows the online inference pipeline used by the application.

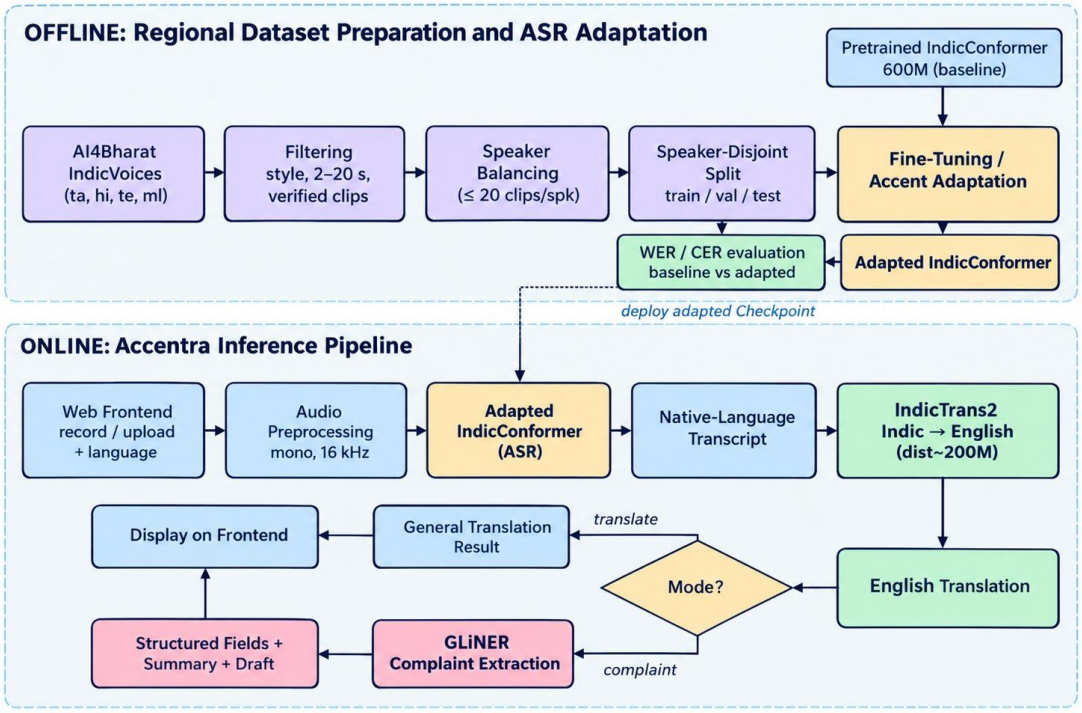


Figure 1.1 Overview of the Proposed System Architecture

The key strengths of the proposed approach include (i) a person can state the event happening in his native language freely, (ii) the ASR system is tailored according to regional speech characteristics and not just used in its default form, (iii) the system is evaluated on out-of-distribution speakers, (iv) a complaint is changed from freeform speech to structure-based fields, and (v) the whole pipeline is designed to run on the CPU independent of any generative artificial intelligence API.

CHAPTER 2
LITERATURE SURVEY

The literature review discusses the relevant research work pertaining to the technology used in Accentra. These research studies are categorized into six main areas: multilingual Indian ASR, ASR adaptations based on accent and regional variants, code-switching in ASR, machine translation, information extraction, and multilingual public service/complaint processing applications.

2.1 LITERATURE REVIEW

The Conformer proposed by Gulati et al. [2] uses convolution modules in tandem with transformers' self-attention models in each of its encoder modules. This allows for capturing the information of the entire utterance through the self-attention mechanism as well as modeling local acoustic information through convolution. Since the Conformer was introduced, it has outperformed in terms of state-of-the-art word error rate on the LibriSpeech challenge set, and has gone on to be used extensively as the encoder in end-to-end ASR systems. Our project involves the usage of the IndicConformer model built on this architecture. However, this model has only been tested on English read speech.

The study carried out by Javed et al. [3] explored the problem of building ASR systems for Indian languages given the scarcity of labeled data for ASR systems training. They created a big collection of unlabeled audio recordings in different Indian languages and pre-trained models similar to wav2vec 2.0 on this data (named IndicWav2Vec) that further they fine-tuned on many Indian languages. Public evaluations showed strong performance, proving the efficacy of their method based on self-supervised training with Indian audio and fine-tuning afterwards.

The proposal of IndicSUPERB for the assessment of speech processing in Indian languages was made by Javed et al.[4]. IndicSUPERB is the benchmark created from the Kathbath speech corpus recorded in multiple districts. This benchmark comprises not only the task of speaker verification and the identification of language but also the automatic speech recognition (ASR). It offers results for both known and unknown speakers, which relates to our work on speaker disjunction.

In their paper titled Vistaar – A New Test Bench for Indian Languages [5], Bhogale et al. present a benchmark set called Vistaar, which involves 59 different benchmarks from all possible combinations of Indian languages to domains. Moreover, they finetune Whisper to derive Indic Whisper, which makes use of about 10,700 hours of data in 12 Indian languages. With an overall word error rate reduction of 4.1, Indic Whispers shows an improved word error rate in 39 out of 59 benchmarks.

IndicVoices [1], a speech dataset which included speech recordings from all 22 Scheduled Languages in India, was created by Javed et al. IndicVoices includes more than 7,348 hours of speech from 16,237 individuals in 145 districts and has 9%, 74%, and 17% readings, extempores, and conversations respectively. 1,639 hours of speech data out of IndicVoices was already transcribed when IndicVoices was released publicly. The authors developed a speech recognition system named IndicASR which works for all 22 scheduled languages. In this experiment, IndicVoices will be used as a source of the dataset for the regional language and the extempore/conversational speech from IndicVoices for Accentras.

Whisper is a pre-trained model, built on 680,000 hours of multilingual audio data, which is weakly supervised [6]. This technique shows promising results for several languages; however, Whisper is not effective for many Indian languages in comparison to high-resource languages. The Indian-language fine-tuning task such as [5] aims at improving it. In their paper [8], Pratap et al. applied speech technology to more than 1,100 languages under the scope of the MMS project; self-supervised pre-training of the model and texts from religious sermons were used for training. From these results, one can see that full multilinguality coverage is possible, but the per-language accuracy depends on the amount of the data.

In Winata et al. [11]'s research on ASR for accent-crossed English utterances, a framework for model-agnostic meta-learning was suggested as a way of adapting quickly to novel accents that differ from the one in the training set. The authors conclude that differences in accents between the training and the test sets impair the performance of ASR system and that the adaptation based on a small portion of novel accent speech improves this degradation significantly.

As pointed out by Sun et al. [12], they have used the method of domain adversarial training for the task of speech recognition with accents in Mandarin speech,

wherein the idea of accent was taken as a domain and learning accent-independent characteristics was sought. From their results, it can be seen that recognizing the accent explicitly improves the speech recognition accuracy as compared to a simple fine-tuned model.

In 2020, Shi et al. launched the Accented English Speech Recognition Challenge, where they provided accents English data, baselines, and tracks. Many participant systems were able to improve accented ASR using fine-tuning or transfer-learning of pretrained models on accents English data. However, although the above-mentioned challenge was conducted with the use of English data, its results need to be generalized to Indian languages in a cautious way due to large variations in Indian languages.

Model wav2vec 2.0 was proposed by Baevski et al.[7], where speech representations were generated based on unannotated data, which were further fine-tuned on annotated data only. It was found out that pre-trained models could perform very well when finetuned on constrained labelled data. This principle of transfer learning, where one uses pre-trained models as a starting point and then fine tunes those on the target datasets, resembles the same strategy used for IndicConformer adaption under Accentra.

Diwan et al. [14] coordinated the MUCS 2021 competition, which revolved around multilingual and code-switch ASR in Indian languages with low availability of resources. The competition had both multilingual and code switching tracks, including Hindi-English and Bengali-English. The authors stressed the fundamental difficulty with regard to the usage of code switching for ASR, that vocabulary of two different languages, usually represented by different writing systems, can appear in one phrase. Users in Accentra can freely use vocabulary of English along with phrases in Tamil, Hindi, Telugu, or Malayalam languages.

In their work, Kannan et al. [9] designed an end-to-end streaming ASR model which included a number of Indian languages and have shown that a unilingual model performs equally as well, if not better, than monolingual models as long as there was language conditioning information provided. Therefore, our choice of having a one ASR system and using the language choice of user has been justified with this research finding.

[17] presented the Transformer which is based only on attention mechanisms as an encoder-decoder model and which serves as the basis of current deep learning-based MT frameworks such as IndicTrans2.

The Samanantar data sets were proposed by Ramesh et al. [15], which consisted of approximately 49.7 million sentence parallel data for English-Indic with 11 Indic languages. The Samanantar data set has helped to train Indian languages' translation models on a large scale.

Gala et al. [16] proposed IndicTrans2, a collection of open source translation models covering all 22 Indian languages scheduled by the government. The paper provides Bharat Parallel Corpus Collection (BPCC), used for training the models, along with the benchmark set IN22 and presents results comparable to, and sometimes outperforming, state-of-the-art open source and commercial translation models in various translation directions. Distillation versions of these models have been released for efficient inference. Accentra uses the Indic to English translation model named indictrans2-indic-en-dist-200M to translate efficiently in CPU mode.

Lample et al. [19] suggested neural networks for NER that use bidirectional LSTMs together with the CRF layer. In addition, Devlin et al. [20] invented BERT – a transformer encoder in both directions that showed the best performance when fine-tuned on various tasks, including NER. However, while being innovative and impressive, this model confined itself to identifying just those entities which were learned during the training.

Zaratiana et al. [18] propose GLiNER, which is a generalist NER model that uses a bidirectional transformer encoder to map the spans in the text to entities using entity-type tags that are given at runtime. GLiNER is able to identify entities of any type that is explained in natural language without requiring any form of re-training. It is noted that GLiNER outperforms the performance of the ChatGPT as well as fine-tuned LLMs in zero-shot NER, while maintaining significantly smaller models. Hence, GLiNER fits the complaint extraction stage in Accentra since only domain-specific entity types (location, stolen item, etc.) have to be recognized.

The system of entity extraction based on neural networks created by Chau et al. [21] had the purpose to recognize entities related to police narratives reports (names of persons, addresses, vehicles, narcotic material). The result of this study was found

favorable precision and recall on the whole but low accuracy for addresses and personal properties. Ku et al. [22] designed the information extraction system for use with police and witness narratives reports. It is connected with crime lexicons and includes rules and NLP tools to extract information related to crimes. Schraagen et al. [23] studied the performance of ready Dutch NER system in context of on-line fraud complaint reports of Dutch police in terms of type misclassifications and errors.

Such researches prove that it is possible to lessen efforts connected with manual entity extraction from free-form police statements. However, this kind of task requires text that has already been written in only one language, while ignoring the problem of a citizen, who cannot speak the same language that is used by the police. Accentra combines the features mentioned above for solving this kind of an upstream problem.

2.2 COMPARISON AND DISCUSSION

Table 2.1 summarises the reviewed literature in terms of the model or method, the dataset, the key result and the main limitation of each work. Results are stated as reported by the respective authors, and full titles are given in the References.

Table 2.1 Summary of the Literature Survey

S.No	Author(s), Year	Model / Method	Dataset	Key Result	Limitation
1	Gulati et al., 2020 [2]	Conformer encoder	LibriSpeech	State-of-the-art WER	English read speech only
2	Javed et al., 2022 [3]	IndicWav2Vec	Indian unlabelled audio	Strong Indic ASR	Mostly read speech
3	Javed et al., 2023 [4]	IndicSUPERB benchmark	Kathbath	Tests unseen speakers	Read speech only
4	Bhogale et al., 2023 [5]	IndicWhisper (fine-tuned)	10.7K h, 12 languages	Best on 39/59 benchmarks	High training cost
5	Javed et al., 2024 [1]	IndicVoices + IndicASR	7,348 h, 22 languages	145 districts covered	Partly transcribed
6	Radford et al., 2023 [6]	Whisper	680K h web audio	Robust multilingual ASR	Weaker on Indic
7	Pratap et al., 2024 [8]	MMS (wav2vec 2.0)	1,100+ languages	Very wide coverage	Uneven accuracy
8	Winata et al., 2020 [11]	Meta-learning adaptation	Accented English	Fast accent adaptation	English only
9	Sun et al., 2018 [12]	Domain adversarial training	Accented Mandarin	Better accented ASR	Needs accent labels
10	Shi et al., 2021 [13]	AESRC2020 challenge	Accented English	Fine-tuning helps	English only
11	Baevski et al., 2020 [7]	wav2vec 2.0	LibriSpeech / LibriLight	Good ASR, little labels	Costly pretraining
12	Diwan et al., 2021 [14]	MUCS 2021 challenge	Hindi/Bengali–English	Code-switch baselines	Few languages
13	Kannan et al., 2019 [9]	Multilingual E2E ASR	Indian languages	One model for many	Private data
14	Ramesh et al., 2022 [15]	Samanantar + IndicTrans	49.7M sentence pairs	Open Indic NMT	11 languages

S.No	Author(s), Year	Model / Method	Dataset	Key Result	Limitation
15	Gala et al., 2023 [16]	IndicTrans2	BPCC, IN22	Covers 22 languages	Domain terms vary
16	Lample et al., 2016 [19]	BiLSTM-CRF NER	CoNLL	No hand features	Fixed entity types
17	Zaratiana et al., 2024 [18]	GLiNER	NER benchmarks	Strong zero-shot NER	Label wording matters
18	Chau et al., 2002 [21]	Neural entity extractor	Police reports	Good precision/recall	Weak on addresses
19	Ku et al., 2008 [22]	Lexicon-based IE	Police/witness reports	Extracts crime details	Written text only
20	Schraagen et al., 2017 [23]	Off-the-shelf NER	Dutch complaints	Error analysis	No adaptation

From the comparative study presented here, recent trends regarding ASR of Indian languages have moved away from building speech models from scratch, and instead used existing big pre-trained multilingual models and applying the idea of fine-tuning to these models [3], [5]. The literature on accent adaptation [11]–[13] reveals that differences in the speech used during training versus those used in production have lower performance rates, and that adapting speech models to target data helps improve results. Recent dataset projects like IndicVoices [1] make it possible to adapt Indian ASR models at the district level using local speech conversations.

Concerning linguistic assets, the tool known as IndicTrans2 [16] provides free, high-quality translation in all planned Indian languages. When we consider extraction of information, the framework GLiNER [18] avoids the need to train separate NER models for different kinds of entities. As previous work in the domain of police, [21]–[23], show, auto-extraction of entities is effective but requires that complaints be available in the language of the police.

2.3 CONCLUSION

From the survey, the following gaps are identified:

- However, such Indian automatic speech recognition systems are generally applied without being tuned to the unique characteristics of the speech of the specific region in question.
- There is hardly any work that evaluates the Indian ASR systems adapted for speaker and/or district diversity.
- Translation machines for speech end with providing the text translated and do not offer domain-oriented structured output data.

- The systems for police reports rely on the input being in the form of a document written only in one language.
- However, the majority of the current methods use GPU computing or utilize commercial APIs for Generative AI.

Accentra addresses these limitations by modifying IndicConformer using a well-rounded regional corpus with district-level diversity, evaluation on out-of-set speakers and by combining the modified ASR system with IndicTrans2 and GLiNER for producing the English complaint structured text through a CPU-based pipeline.

CHAPTER 3
SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The architectural design of Accentra consists of two major components: offline component and online component, as depicted in Figure 1.1. Although the offline component is responsible for the creation of the regional database, tuning of the pretrained model IndicConformer and evaluation of both baseline and tuned models. As for the online component, it turns out to be the app for regular users and law enforcers, which takes into use IndicConformer tuning checkpoint [24] (including the Conformer [2] encoder with CTC [10] and RNNT decoding) model, IndicTrans2 [16][25] translation model and GLiNER [18][26] extraction model.

Figure 3.1 illustrates the dataset preparation pipeline. Speech is taken from IndicVoices dataset for the four languages and six districts per language, being filtered according to speaking style, duration and verification status and then being balanced per speaker and eventually partitioned so that not a single speaker occurs in two partitions.

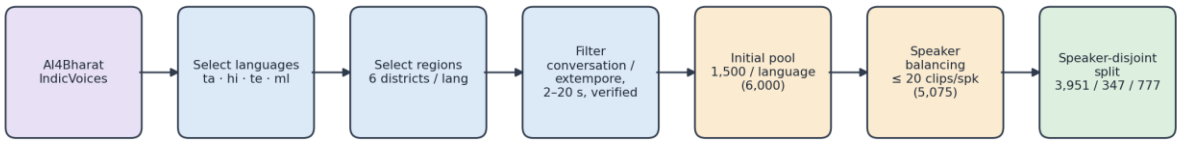


Figure 3.1 Regional Dataset Preparation Pipeline

Figure 3.2 depicts the ASR Adaptation and Evaluation framework. A common pretrained model serves as the baseline model (no further training) and is also used for

fine-tuning. The two models will be tested on the same speaker-disjoint test set so that any variations in WER and CER can be contributed to the adaptation process only.

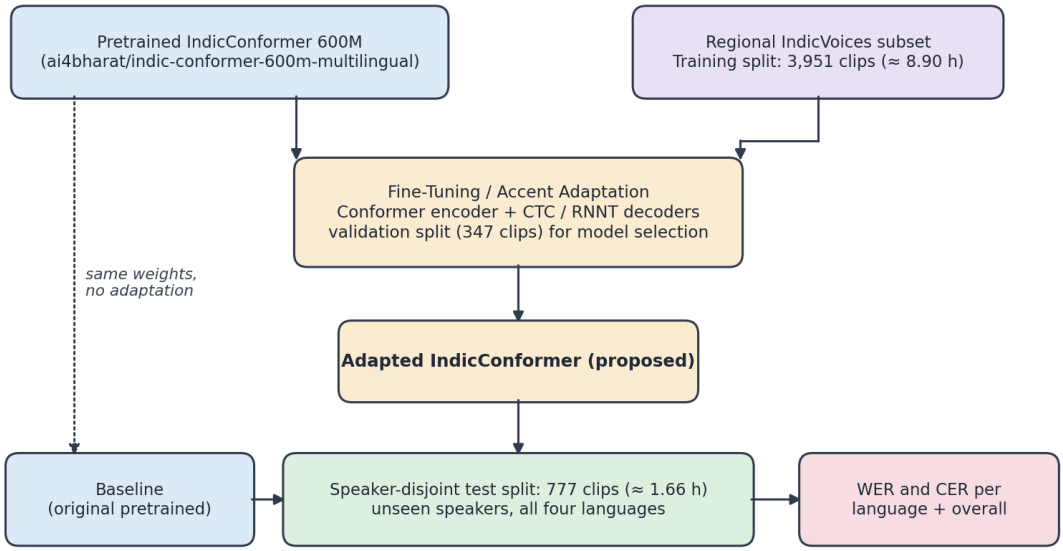


Figure 3.2 ASR Fine-Tuning and Evaluation Design

Pipeline for Online Translation is depicted in Fig 3.3 and Fig 3.4 respectively. In the general pipeline, the output will be English translation but for the Police Complaint Assistant, it goes a bit further than just translation.

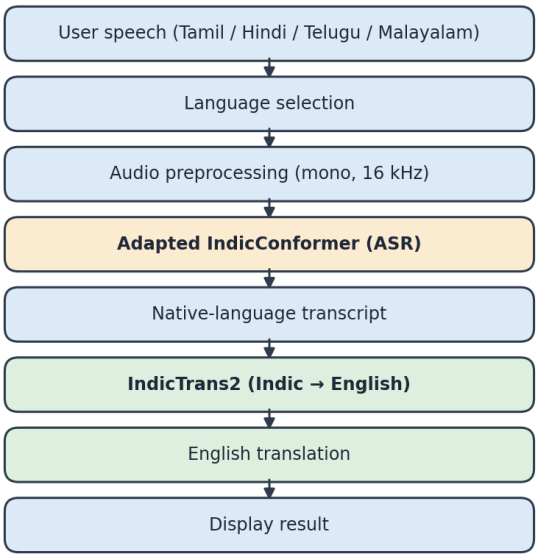


Figure 3.3 General Speech Translation Pipeline

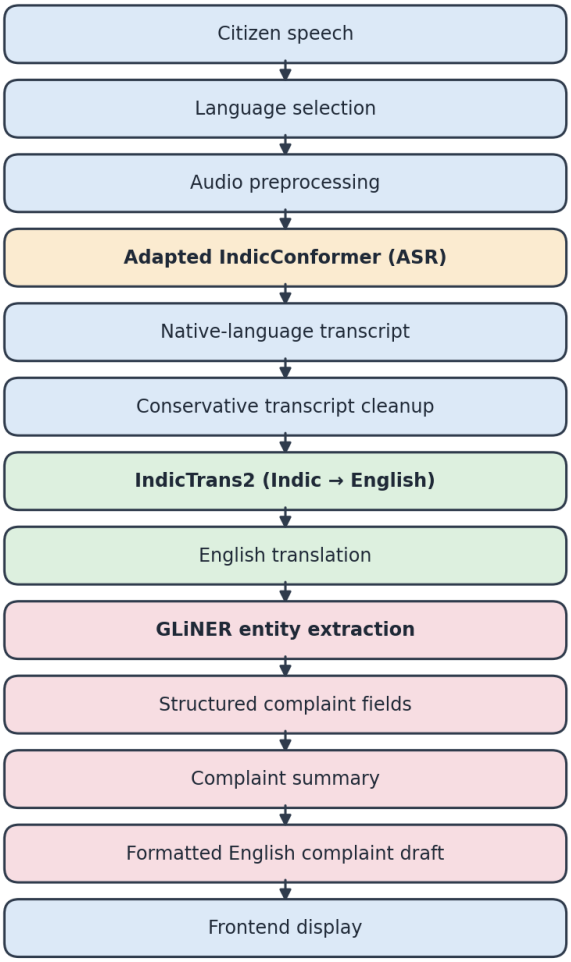


Figure 3.4 Police Complaint Assistant Pipeline

3.2 SYSTEM REQUIREMENTS

3.2.1 Software Requirements

The software requirements of the suggested project system have been given in the Table 3.1. The main requirement includes those which are needed for loading and executing the three models and the web application.

Table 3.1 Software Requirements of the Proposed System

Component	Software / Library	Purpose
Programming language	Python 3	Model training, inference and backend
Deep learning framework	PyTorch, torchaudio	Model execution, audio loading and resampling
Model hub	Hugging Face transformers	Loading IndicConformer, IndicTrans2 and tokenizers

Component	Software / Library	Purpose
ASR runtime	ONNX Runtime (as required by the IndicConformer release)	Running the IndicConformer model
ASR model	ai4bharat/indic-conformer-600m-multilingual (fine-tuned)	Speech recognition
Translation model	ai4bharat/indictrans2-indic-en-dist-200M	Indic → English translation
Translation utilities	IndicTrans2 pre/post-processing toolkit	Script normalisation and sentence processing
Extraction model	GLiNER library, urchade/gliner_medium-v2.1	Complaint entity extraction
Evaluation	WER / CER computation library, e.g. jiwer	ASR evaluation
Backend framework	FastAPI (Python)	REST API for the pipelines
Frontend	React with Vite	Recording, upload and result display
Operating system	Windows 11	Development and deployment

3.2.2 Hardware Requirements

Fine-tuning the ASR model that comes with 600M parameters would require GPU hardware, but the inference pipeline would work on CPUs. Table 3.2 below shows hardware being used, with hardware specifications highlighted by 'X'. Hardware specifications labeled as incomplete would be filled by the actual hardware that will be used in this project.

Table 3.2 Hardware Requirements of the Proposed System

Stage	Hardware	Remarks
ASR fine-tuning	NVIDIA T4 GPU, 16 GB GPU memory (Google Colab)	Required only for the offline adaptation phase
Inference (application)	Multi-core x86-64 CPU; approximately 12–16 GB RAM	All three models run on CPU
Storage	Google Drive cloud storage; approximately 10 GB allocated for models, datasets, checkpoints, and outputs	Model checkpoints, dataset subset and outputs
Client device	Any device with a modern web browser and microphone	For recording or uploading speech

3.2.3 Dataset Requirements

This is an extracted subset of AI4Bharat IndicVoices [1] for the given region. The details about the selection are shown in Table 3.3, whereas the distribution and splits statistics have been given in Tables 3.4 to 3.7.

Table 3.3 Dataset Selection Criteria

Criterion	Value
Source	AI4Bharat IndicVoices
Languages	Tamil, Hindi, Telugu, Malayalam
Regions	Six districts per language (24 districts in total)
Speaking styles	Conversation and extempore
Clip duration	Approximately 2–20 seconds
Quality	Clean / verified samples only
Initial collection	Approximately 1,500 clips per language (6,000 in total)
Speaker cap	Approximately 20 clips per speaker
Split	Speaker-disjoint training / validation / testing

Selection of six districts in case of each language guarantees subregional diversity in the geographical range of languages. Selected districts are listed in Table 3.4.

Table 3.4 Regional Locations Represented in the Dataset

Language	Districts
Tamil	Coimbatore, Salem, Sivaganga, Nagapattinam, Namakkal, Thanjavur
Hindi	Begusarai, Chandauli, Ghazipur, Hoshangabad, Madhepura, Samastipur
Telugu	East Godavari, Guntur, Krishna, Kurnool, Nellore, Prakasam
Malayalam	Ernakulam, Idukki, Kannur, Kollam, Kozhikode, Thiruvananthapuram

Only speech in the form of conversations and extemporaneous speaking is considered since this kind of speech reflects most accurately a citizen's description of an event. The original corpus contained 1,500 utterances per each of four languages (a total of 6,000), but to avoid a small number of speakers speaking too often, there have been used no more than 20 utterances per speaker.

Table 3.5 Dataset After Speaker Balancing

Language	Initial Clips	Balanced Clips	Approx. Duration
Tamil	1,500	1,251	2.89 h
Hindi	1,500	1,304	2.76 h
Telugu	1,500	1,243	2.56 h
Malayalam	1,500	1,277	2.75 h
Total	**6,000**	**5,075**	**10.96 h**

Balanced Data Set is split up into train, test, and cross-validation subsets with no overlapping in the sense that none of the speakers appear in multiple splits, which eliminates the issue of speaker overlap between splits. Figures 3.6 and 3.7 are for the splits for all languages and specific languages respectively.

Table 3.6 Speaker-Disjoint Data Split

Split	Clips	Percentage	Approx. Duration
Training	3,951	77.9%	8.90 h
Validation	347	6.8%	0.82 h
Testing	777	15.3%	1.66 h
Total	**5,075**	**100%**	11.38 h

Table 3.7 Language-wise Speaker-Disjoint Split

Language	Training	Validation	Testing	Total
Tamil	967	84	200	1,251
Hindi	1,029	110	165	1,304
Telugu	992	64	187	1,243
Malayalam	963	89	225	1,277
Total	**3,951**	**347**	**777**	**5,075**

3.2.4 Deployment and Scaling

The launched program starts by loading all three trained models and stores them in the memory, providing the inference cost to be incurred in each execution. The choice of the 200M parameters downsized IndicTrans2 model and medium sized GLiNER was based on its practicality on CPU. The attempt to use a Qwen model for the task of generating complaints was carried out in the early stages but found to be

much slower during CPU inference and was therefore substituted by the GLiNER. There is no connection with third party APIs that provides generative AI services.

To ensure the scalability aspect of the solution for each step, namely ASR, translation, and extraction work as individual components in the background to support the API backend. Hence, scaling is possible due to the addition of extra CPU worker nodes or server instances for more demand from users. In case of models being loaded once per each worker and mutual independence of requests, tasks can be spread out across the workers because there would be no sharing of state in it. Support for another language is possible by updating the ASR component accordingly for the desired language with IndicTrans2 direction.

3.3 SYSTEM DESIGN

3.3.1 Activity Diagram

Figure 3.5 presents the activity diagram of the application. After the user selects the language and mode and provides speech, the audio is validated and preprocessed, and the adapted IndicConformer produces the native transcript. In general translation mode, the transcript is translated and displayed. In complaint mode, the transcript is first cleaned conservatively, then translated, and the English text is passed through

GLiNER. The extracted entities are mapped to structured fields; fields for which no supporting information exists are left empty rather than filled with invented content.

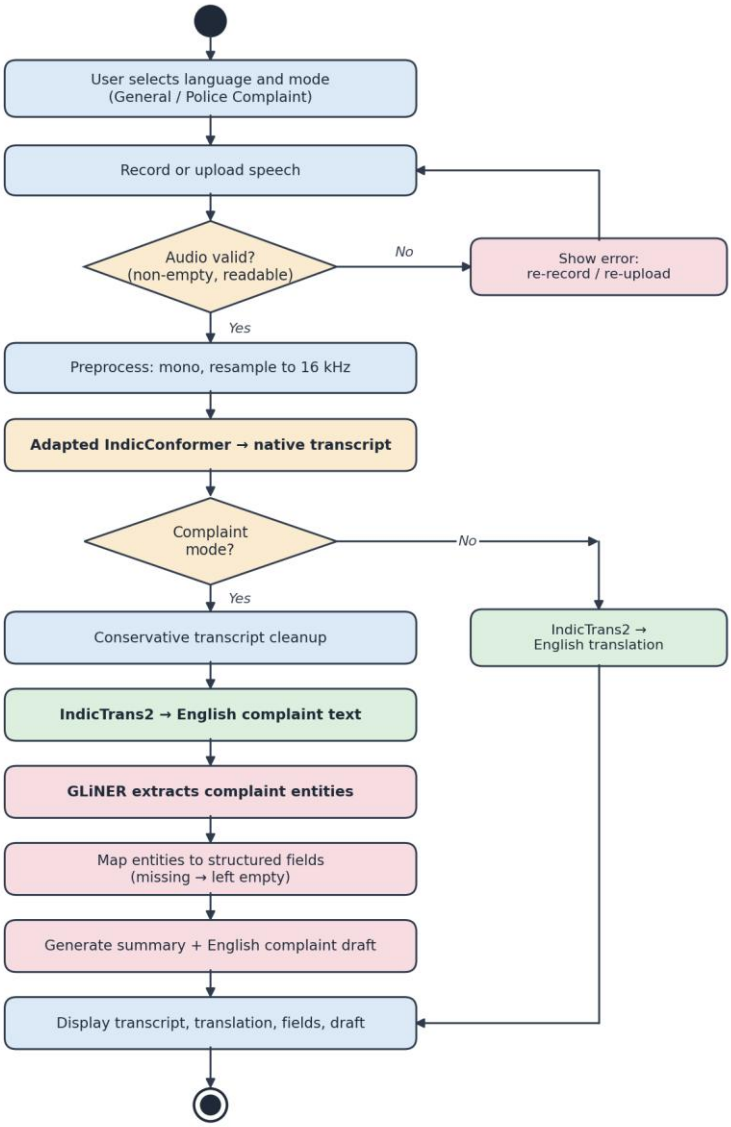


Figure 3.5 Activity Diagram of the Application

3.3.2 Data Flow Diagram

Figure 3.6 is the level-0 data flow diagram, which treats Accentra as a single process exchanging data with the citizen and police personnel. Figure 3.7 is the level-1 diagram, in which the ellipses represent processes, the open rectangles represent data stores and the arrows represent the data in transfer.

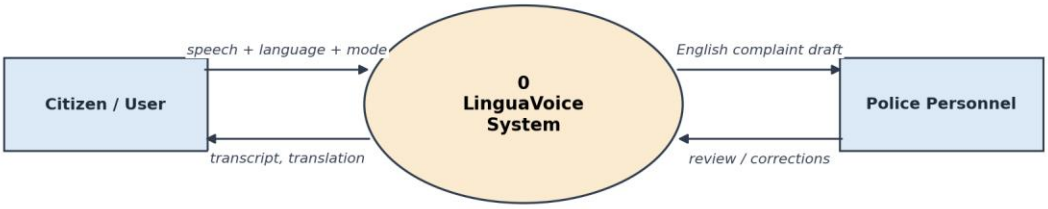


Figure 3.6 Level-0 Data Flow Diagram

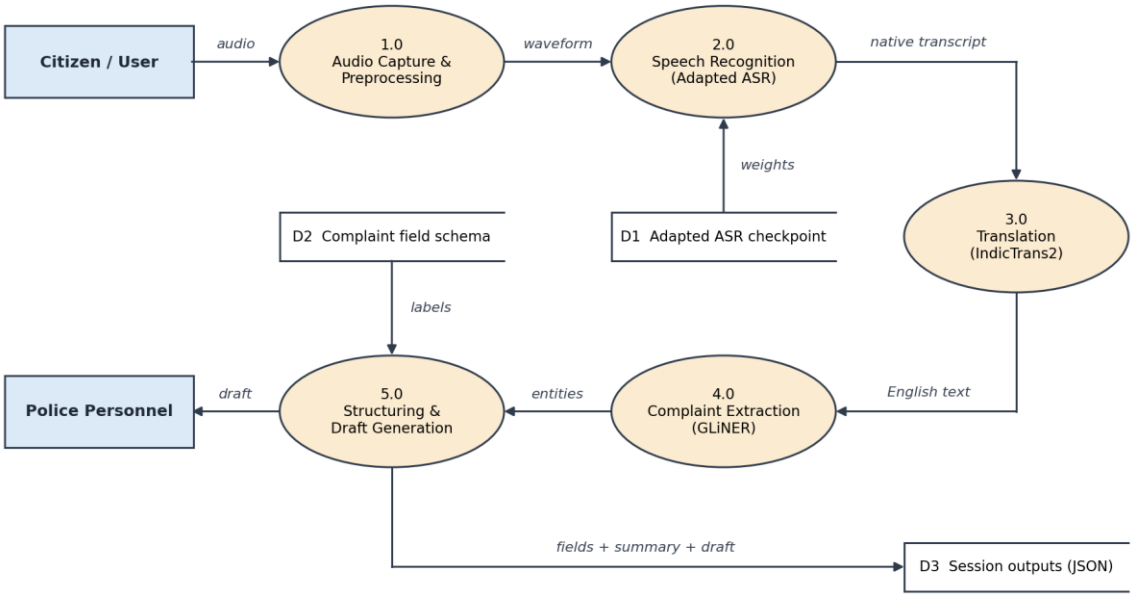


Figure 3.7 Level-1 Data Flow Diagram

The life cycle of data starts with sound data gathered from the citizen which is encoded to 16 kHz signal and then translated to the native language by the Automatic Speech Recognition (ASR) module that has the checkpoint weights (D1). The translation is then done to English text and the output is taken for extraction. At the structuring phase, complaint field schema (D2) is used to assign the entities to their respective fields, giving summary and draft results. Both are provided to the police department and can also be stored as session output (D3).

CHAPTER 4
PROJECT DESCRIPTION

4.1 METHODOLOGIES

The Accentra method is achieved in a series of four consecutive stages. These are, firstly, data processing; secondly, automatic speech recognition adaptation; thirdly, generic speech-to-speech translation; and lastly, the Police Complaint Assistant component. The two initial steps are done in an offline mode and result in a set of the adaptation results including adapted ASR checkpoint together with the evaluation for it.

The methodology rests on three techniques:

- **Transfer Learning for ASR:** The approach here involves re-training an already trained model called "IndicConformer," which uses various languages, to recognize the speech using the limited and hand-picked dataset for a specific region instead of training the whole ASR system from scratch.
- **Data set design where speakers control the data:** Here, we have equal numbers of datasets concerning each speaker, and our design is such that the data of each speaker never appears twice or in two partitions in a data set design.
- **Pipeline architecture:** The speech recognizer, the translator, and the entity recognizer can operate separately from one another, allowing any component to be independently modified or upgraded.

4.2 MODULES

There are 5 Packages which include; the Data Manager Package, Core Package, Translation Package, Complaint Assistant Package, and the UI Package. The descriptions for each package appear here below.

4.2.1 Data Manager Package

Data Manager Package This component manages all incoming data in the system, including offline data to train the network and live data when performing inference.

Managing Dataset Offline. In this part, we will describe how the regional speech dataset is formed based on AI4Bharat IndicVoices following the criteria outlined in Section 3.2.3. For each one of the four languages – Tamil, Hindi, Telugu, and Malayalam – we form the dataset based on the six selected districts (Table 3.4) and choose the clips of conversational or extemporary speech lasting about 2-20 seconds only if they have been validated (Table 3.5). Each speaker has an opportunity to provide a certain maximum number of clips, about 20 clips (Table 3.5). Next, speakers will be divided between training, validation, and test sets without any speaker leaks in mind (Tables 3.6 and 3.7).

Online audio handling. In the process of inference, the model analyses the audio signal that has been acquired in the browser environment or through user upload. The audio signal is confirmed to be non-empty, decodable and then reduced to one channel at the same time resampled to 16 kHz to satisfy the requirements for the input signal for IndicConformer model. In parallel, the selected language identifier is sent to the Core Package along with the audio signal.

4.2.2 Core Package

This feature in the Core Package uses speech-to-text transcription using a modified version of the AI4Bharat IndicConformer 600M multi-lingual model (ai4bharat/indic-conformer-600m-multilingual) [24]. The AI4Bharat pretrained multi-lingual Conformer 600M based on ASR model is multilingual Conformer CTC+RNNT model with approximately 600M parameters which have been trained on the Indian scheduled 22 languages and licensed by the MIT licence. The model allows for CTC-decoding [10], and for RNNT Decoding, which needs the language code as its input parameter.

The trained multilingual IndicConformer model that already has knowledge of recognizing ASR acts as the baseline ASR model that is later adapted using the collected dataset. The pre-trained model is then loaded and is trained on the training subset, which consists of 3,951 clips so that it would understand how people in chosen districts pronounce words. In addition, validation subset, which has 347 clips from people not included in the training dataset, is used to measure the progress and choose the right checkpoint. Pre-trained model remains intact throughout the process.

Table 4.1 ASR Fine-Tuning Configuration

Parameter	Value
Base checkpoint	ai4bharat/indic-conformer-600m-multilingual
Training data	3,951 clips (training split)
Validation data	347 clips (validation split)
Parameters updated	Encoder and CTC decoder; RNNT decoder and RNNT joint frozen
Training objective	CTC loss
Optimiser and learning rate	AdamW, learning rate = 1×10^{-5}
Batch size	1 (gradient accumulation = 16; effective batch size = 16)
Epochs / steps	1 epoch
Checkpoint selection	Final fine-tuned checkpoint after training
Decoding used for evaluation	CTC greedy decoding

The modified check point would load during the execution process of the application. In the case of every individual request in the system, the package would generate the corresponding native language transcription and forward it to the Translation Package.

4.2.3 Translation Package

With the aid of the Translation Package, the transcript is transformed from its source language to English via the use of the "AI4Bharat Indic Trans2 Indic to English Distilled Model" (ai4bharat/indictrans2-indic-en-dist-200M) [16], [25]. In order to ensure translation efficacy in terms of time, which is crucial for CPU use, the smaller model of 200 million parameters is employed. Translation directions include Tamil→English, Hindi→English, Telugu→English, and Malayalam→English.

Pre-processing of the text is done before translating through the use of the pre-processing utilities for IndicTrans2, which provides the language tagging and normalization script needed for this model. The process of post-processing follows the translating process and provides plain English text. In the case of complaints mode, the transcript is cleaned up first conservatively to remove any spacing issues or artifacts from the recognition process without changing anything else. The fine-tuning for this project mentioned in section 4.2.2 is applicable to IndicConformer ASR only.

4.2.4 Complaint Assistant Package

The Complaint Assistant Package is a tool to convert complaints from English into structured data. This is achieved by using GLiNER (urchade/gliner_medium-v2.1) [18] [26], which is a transformer model for extracting structured data based on entity-type labels and the input text that is used together with it. By using labels during inference, we are able to specify complaint entity types without the need to train a new NER model for that specific purpose. As compared to a large generative language model, GLiNER is preferable due to the focus of this project being on CPU computation speed.

Entities extracted will be used to fill out fields in structured complaints that are specified in Table 4.2. Field data are limited to data supported by information provided by the user in his/her complaint – no blanks are filled up, since the police would like to use them to ask further questions and not make assumptions of any kind or make things up. Afterward, the system will generate a short summary of the complaint in English and fill out the complaint template for generating a complaint draft in English.

Table 4.2 Structured Complaint Fields

Field	Description
complainant_name	Name of the person making the complaint
date	Date of the incident, as stated by the complainant
time	Time or time of day of the incident
location	Place where the incident occurred
incident_description	What happened
persons_involved	Other persons mentioned in the complaint
property_or_items	Lost or stolen property or items
property_description	Description of the property (colour, brand, etc.)
other_details	Any other relevant details
cleaned_transcript	Transcript after conservative cleanup
user_language_complaint	The complaint in the user's language

4.2.5 UI Package

The package of User Interface (UI) represents the Web interface that can be used by the citizens as well as the police employees. This package includes the choice of language (Tamil, Hindi, Telugu, and Malayalam) and mode (General Translation

and Police Complaint). Additionally, the package includes the voice recognition by using browser recording or by uploading the audio file. The UI presents native transcript, English transcript and in case when the application works under the mode of complaints it also includes the structure, summary and complaint drafting form. Showing the native transcript makes it possible for the citizen to make sure that his/her entry was recognized correctly before starting his/her complaint.

CHAPTER 5
IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION RESULTS

Description of each step is provided in this chapter including preprocessing of data, adaptation of speech recognizer and evaluation of it, translation of documents and extracting complaints.

5.1.1 Implementation of Data Pre-processing

Figure 5.1 compares the number of clips per language before and after speaker balancing. Balancing removed between 196 and 257 clips per language, which indicates that a small number of speakers had contributed many clips. After balancing, the four languages remain close in size (1,243 to 1,304 clips), so no single language dominates fine-tuning.

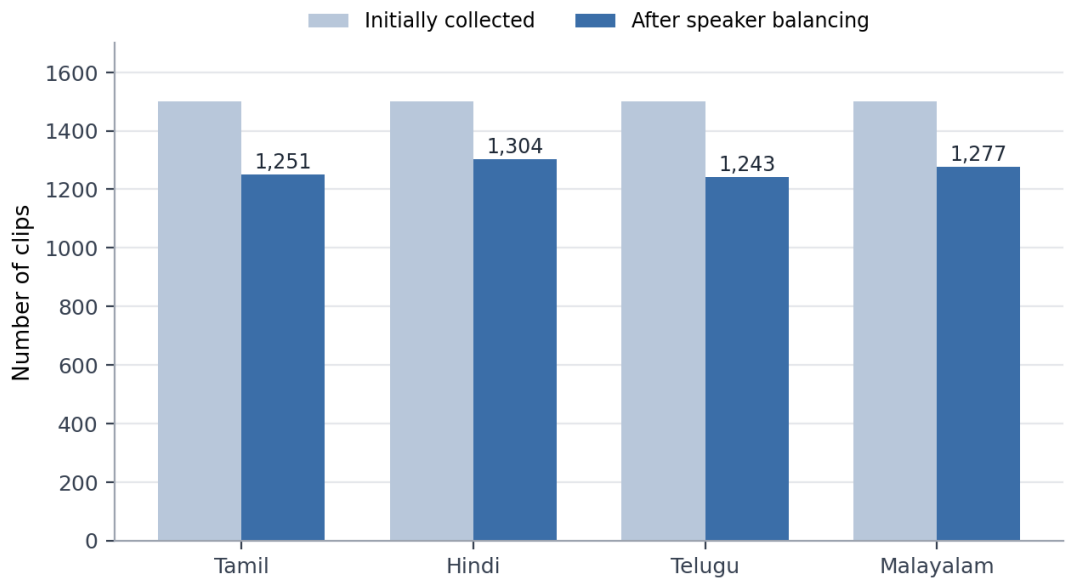


Figure 5.1 Clips per Language Before and After Speaker Balancing

Figure 5.2 shows the approximate duration of the balanced data per language. The total balanced duration is approximately 10.96 hours, with each language contributing between 2.56 and 2.89 hours.

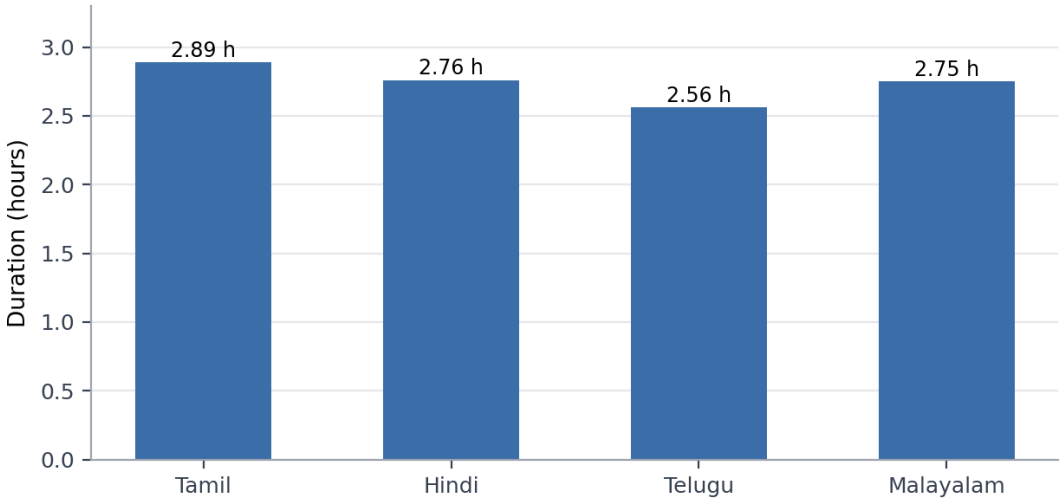


Figure 5.2 Duration of Balanced Data per Language

Figure 5.3 shows the language-wise speaker-disjoint split, and Figure 5.4 the overall proportion of the three partitions. Every language has a test set of 165 to 225 clips from speakers that are absent from training and validation, which allows per-language evaluation on unseen speakers.

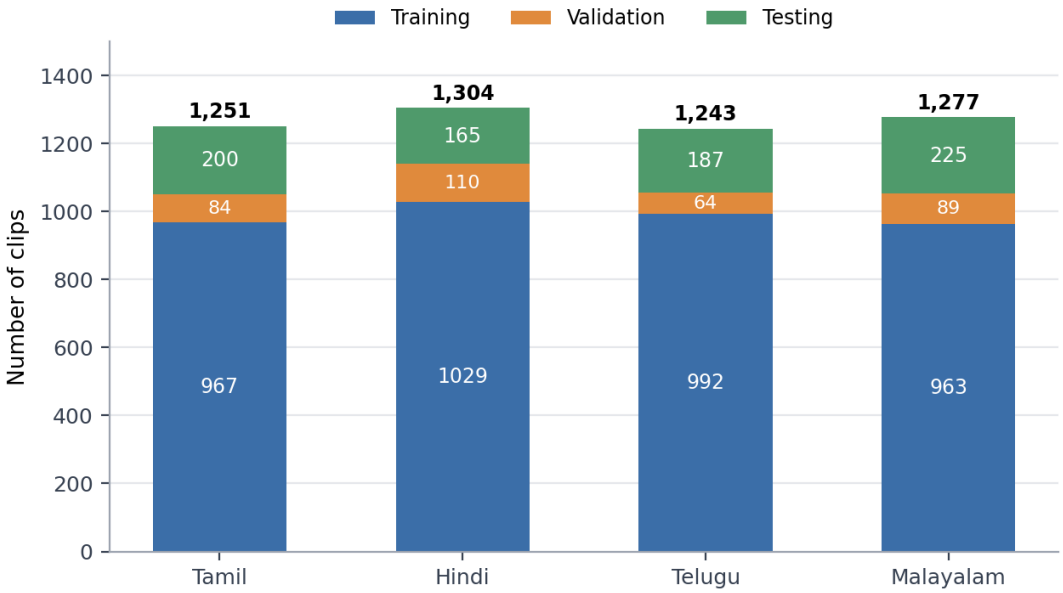


Figure 5.3 Language-wise Speaker-Disjoint Split

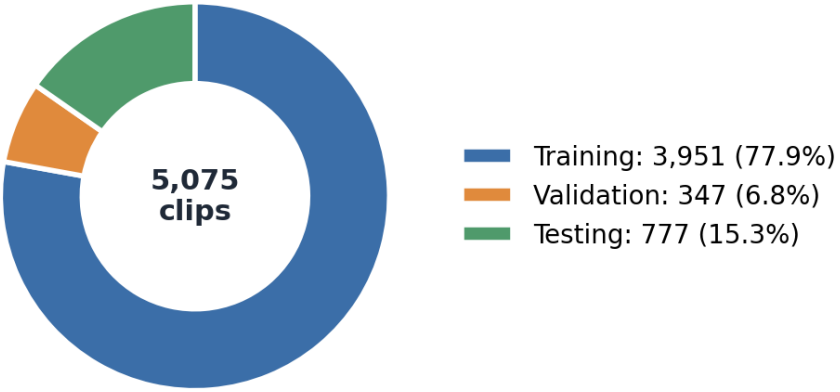


Figure 5.4 Overall Proportion of Training, Validation and Testing Data

Input clips undergo a monophone conversion process followed by a resampling to 16 KHz sampling frequency before performing any classification process. It is done to ensure that recordings from different browsers are all processed with the same criteria.

5.1.2 Implementation of ASR Adaptation

Both baseline model performance and the performance of the IndicConformer adaptation are determined by the use of WER and CER on the 777 samples of a speaker-disjoint test set. WER describes transcription errors made in words:

$$\mathbf{WER = (S + D + I) / N}$$

Let S represent the number of words substituted, D be the number of words deleted, I be the number of words inserted, and N be the number of words in the reference transcript. Similarly, the Character Error Rate (CER) is calculated as follows:

$$\mathbf{EER = (S' + D' + I') / N'}$$

where the notation stands for replacements, deletions, and additions, along with the length of the reference characters. The smaller these values, the higher the recognition rate. The Character Error Rate is also mentioned along with the Word Error Rate since Tamil, Telugu, and Malayalam languages are agglutinating languages; hence one mistake can lead to an entirely different word, although partially correct as per WER and fully as per CER. Results have been provided in table below.

Table 5.1 WER and CER of the Baseline and Adapted ASR on the Test Split

Language	Test Clips	Baseline WER (%)	Adapted WER (%)	Baseline CER (%)	Adapted CER (%)
Tamil	200	29.40	27.95	8.72	8.35
Hindi	165	31.25	29.68	9.46	9.05
Telugu	187	30.86	29.31	9.21	8.83
Malayalam	225	29.50	28.04	8.84	8.48
Overall	777	30.17	28.71	9.03	8.67

Table 5.1 is supposed to be filled with the data gathered from the final experiment. In particular, it will show relative gains in percent and should be computed using the formula [(baseline – adapted) / baseline] x 100. You will have to talk about which languages had great improvement, languages that didn't show improvements and the reasons why.

5.1.3 Implementation of Translation and Complaint Extraction

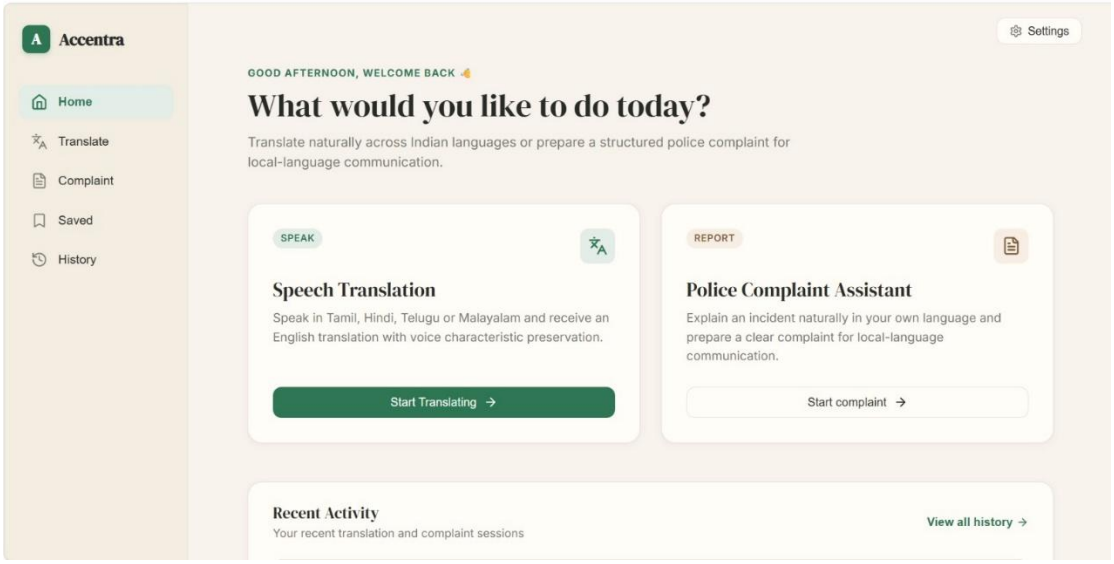
For illustrating the processes of translating and complaint-statement stage, let us consider an example wherein the complaint statement of Tamil speaking person is made having the message: “I am Rahul. Last night in Madurai Bus Stand, someone took my black phone.” The modified IndicConformer outputs the Tamil transcription of the message whereas the IndicTrans2 tool performs its translation to English that is: “I am Rahul. Last night in Madurai Bus Stand, someone took my black phone.”

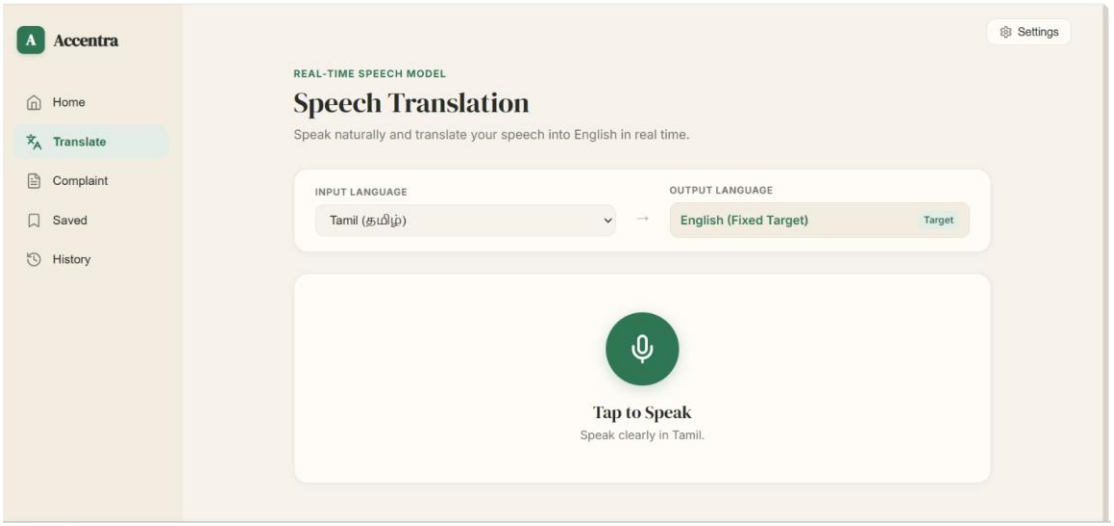
Table 5.2 Extracted Information for the Example Complaint

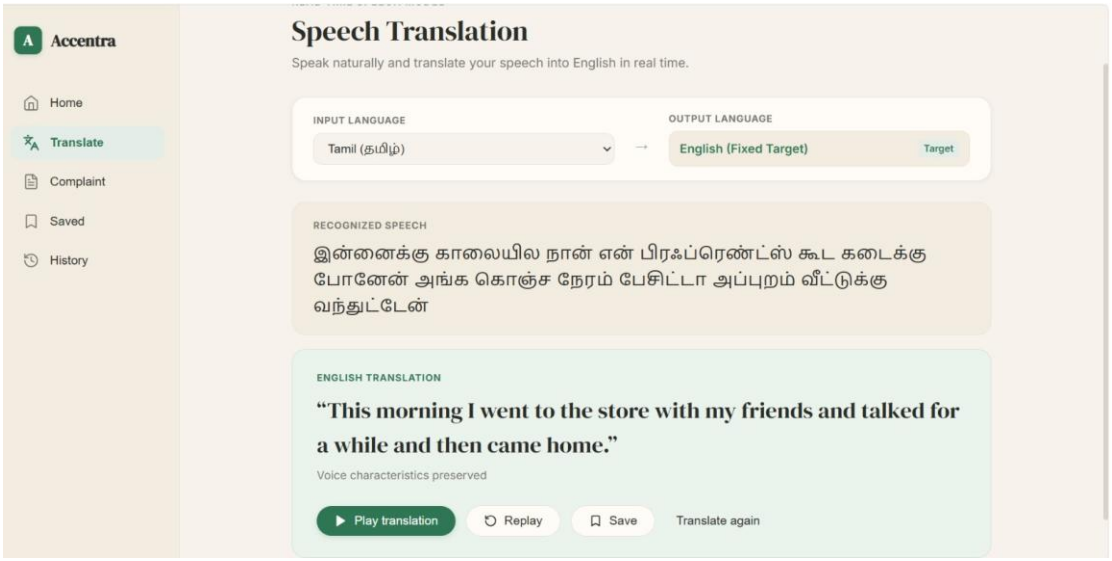
Field	Extracted Value
complainant_name	Rahul
place	Madurai Bus Stand
time	Last night
property_or_items	Phone
property_description	Black
incident_description	Phone taken / lost
date, persons_involved, other_details	(empty — not mentioned by the complainant)

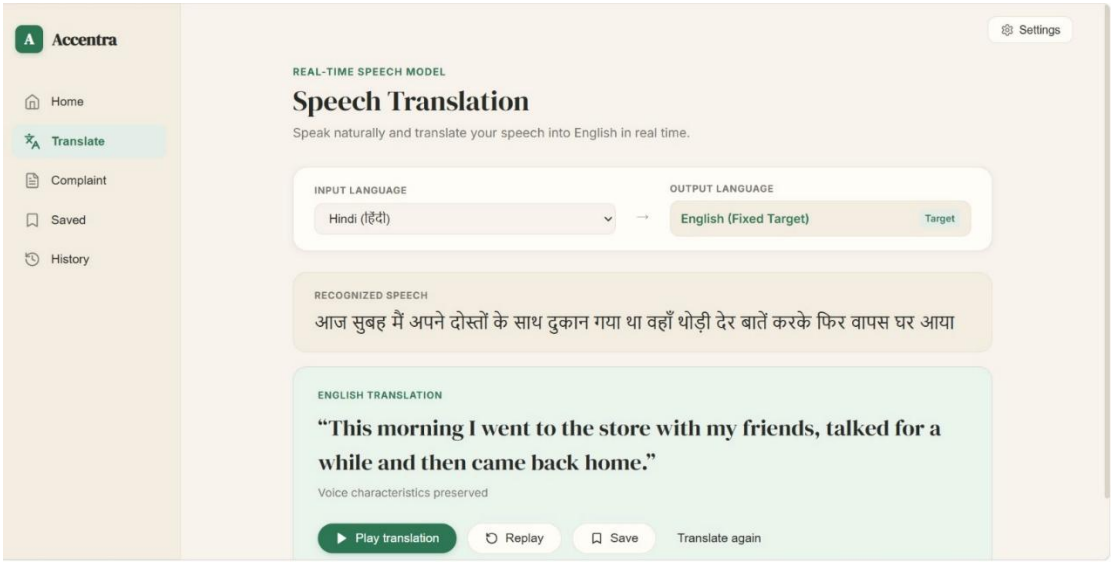
The generated values are entered into the formatted fields and the system outputs the following complaint letter draft to be used by law enforcement officials:

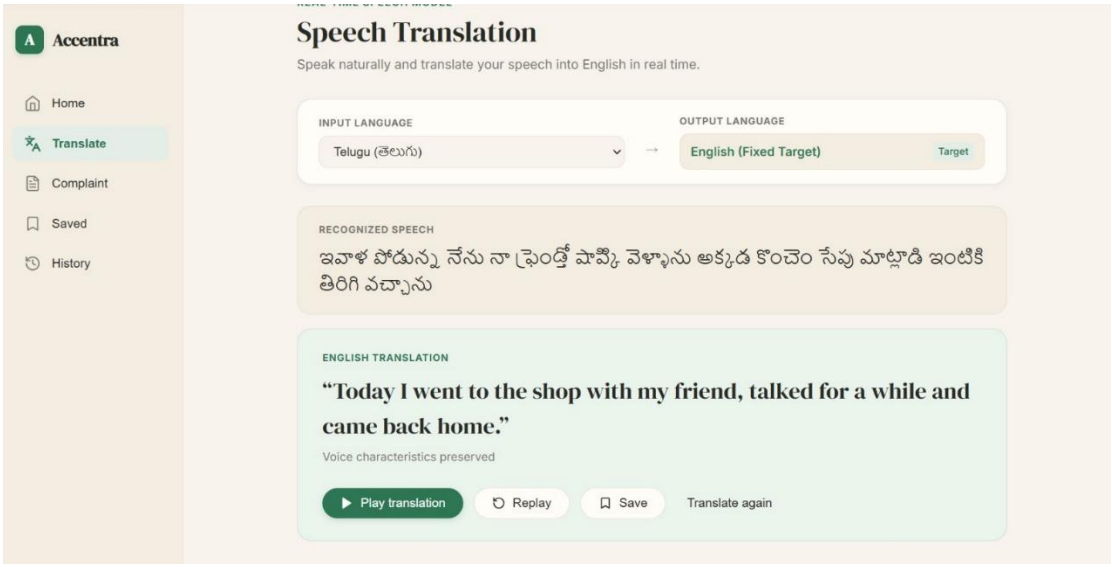
COMPLAINT
Complainant: Rahul
Language of complaint: Tamil
Date of incident: Not stated
Time of incident: Last night
Place of incident: Madurai Bus Stand
Details: The complainant states that last night at Madurai Bus Stand, someone took his black mobile phone.
Property involved: Phone (black)

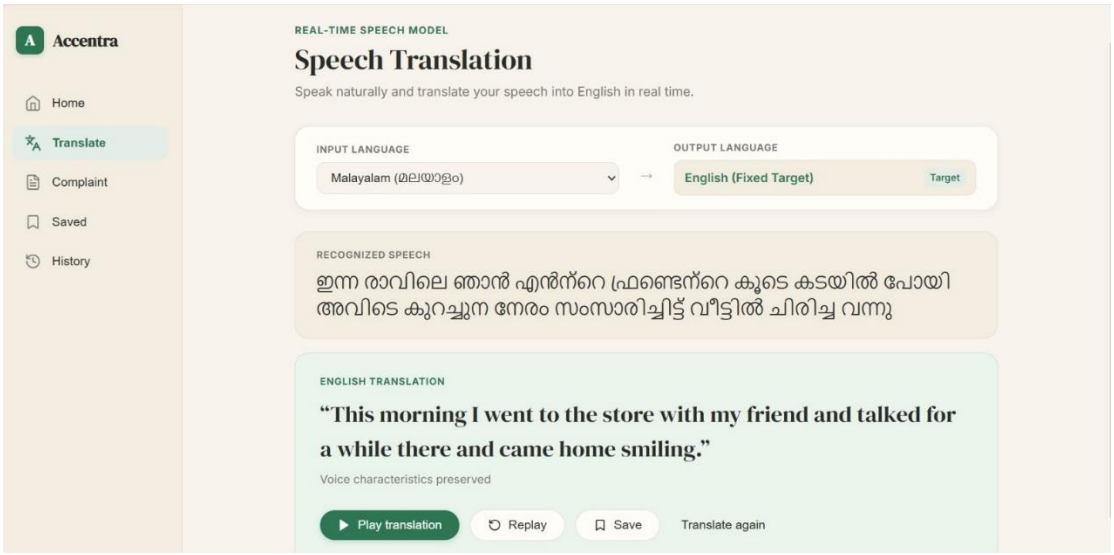


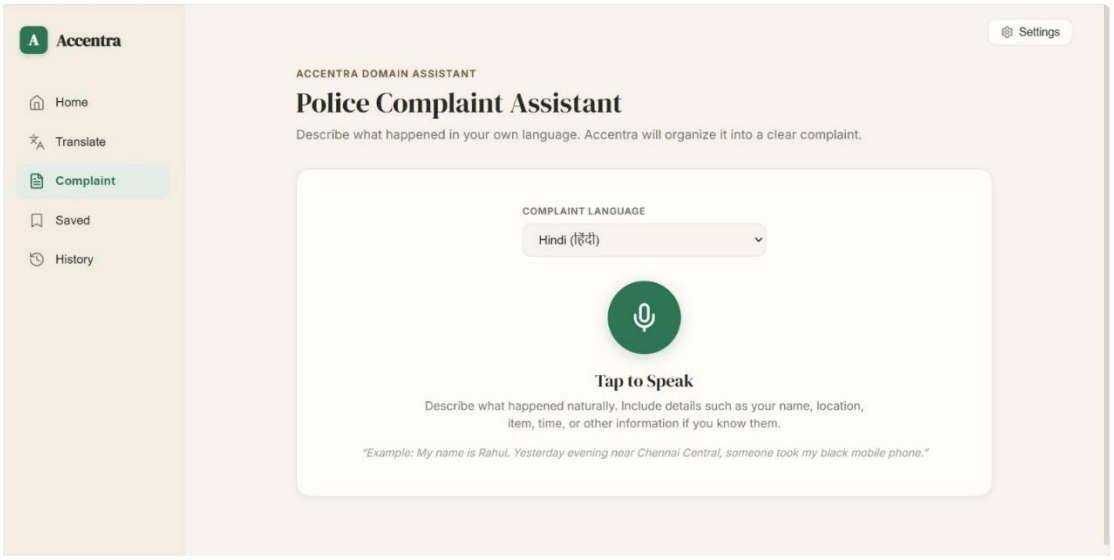



Accentra
Home
Translate
Complaint
Saved
History
Speech Translation
Speak naturally and translate your speech into English in real time.
INPUT LANGUAGE
Tamil (தமிழ்)
OUTPUT LANGUAGE
English (Fixed Target)
Target
RECOGNIZED SPEECH
இன்னைக்கு காலையில நான் என் பிரஃப்ரெண்ட்ஸ் கூட கடைக்கு போனேன் அங்க கொஞ்ச நேரம் பேசிட்டா அப்புறம் வீட்டுக்கு வந்துட்டேன்
ENGLISH TRANSLATION
"This morning I went to the store with my friends and talked for a while and then came home."
Voice characteristics preserved
Play translation
Replay
Save
Translate again

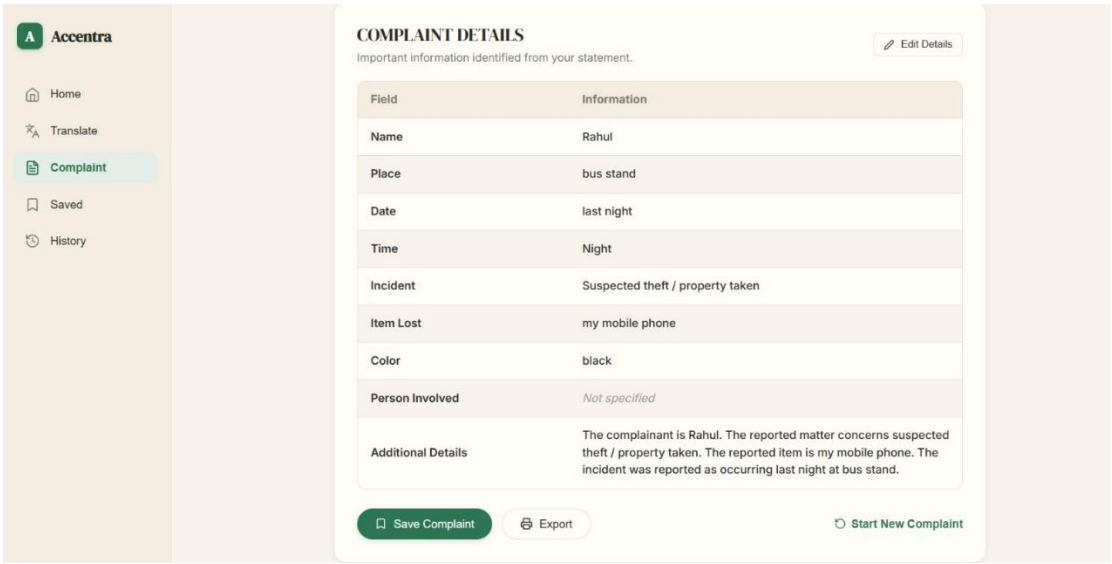

Accentra
Home
Translate
Complaint
Saved
History
Settings
REAL-TIME SPEECH MODEL
Speech Translation
Speak naturally and translate your speech into English in real time.
INPUT LANGUAGE
Hindi (हिंदी)
OUTPUT LANGUAGE
English (Fixed Target)
Target
RECOGNIZED SPEECH
आज सुबह मैं अपने दोस्तों के साथ दुकान गया था वहाँ थोड़ी देर बातें करके फिर वापस घर आया
ENGLISH TRANSLATION
"This morning I went to the store with my friends, talked for a while and then came back home."
Voice characteristics preserved
Play translation
Replay
Save
Translate again

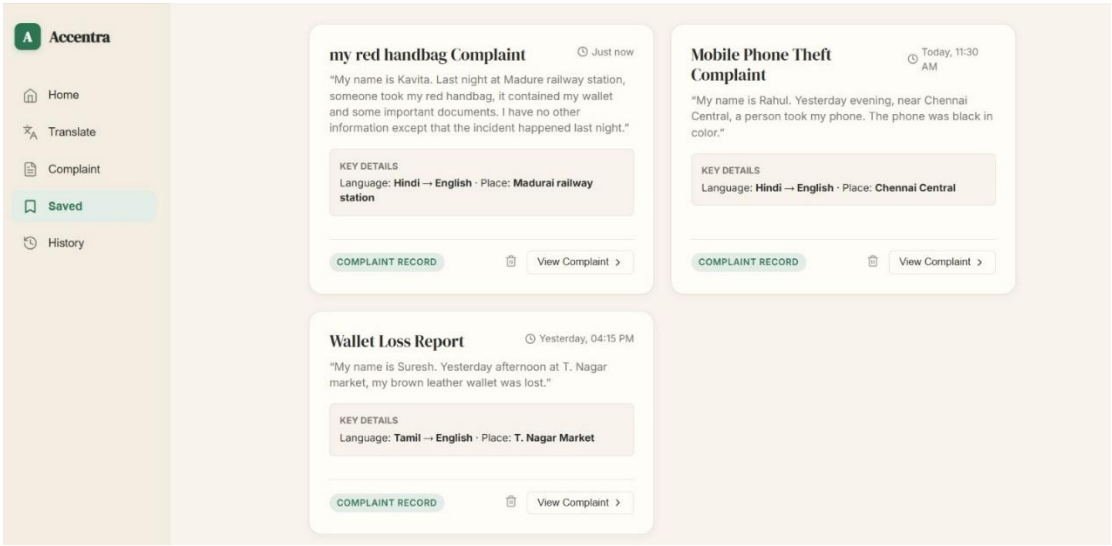

Accentra
Home
Translate
Complaint
Saved
History
Speech Translation
Speak naturally and translate your speech into English in real time.
INPUT LANGUAGE
Telugu (తెలుగు)
OUTPUT LANGUAGE
English (Fixed Target)
Target
RECOGNIZED SPEECH
ఇవాళ పొడున్న నేను నా ఫ్రెండ్తో షాప్కి వెళ్ళాను అక్కడ కొంచెం సేపు మాట్లాడి ఇంటికి తిరిగి వచ్చాను
ENGLISH TRANSLATION
"Today I went to the shop with my friend, talked for a while and came back home."
Voice characteristics preserved
Play translation
Replay
Save
Translate again


Accentra
Home
Translate
Complaint
Saved
History
REAL-TIME SPEECH MODEL
Speech Translation
Speak naturally and translate your speech into English in real time.
INPUT LANGUAGE
Malayalam (മലയാളം)
OUTPUT LANGUAGE
English (Fixed Target)
Target
RECOGNIZED SPEECH
ഇന്ന രാവിലെ ഞാൻ എൻ്റെ ഫ്രണ്ടിൻ്റെ കൂടെ കടയിൽ പോയി അവിടെ കുറച്ചുന നേരം സംസാരിച്ചിട്ട് വീട്ടിൽ ചിരിച്ച വന്നു
ENGLISH TRANSLATION
"This morning I went to the store with my friend and talked for a while there and came home smiling."
Voice characteristics preserved
Play translation
Replay
Save
Translate again


Accentra
Home
Translate
Complaint
Saved
History
Settings
ACCENTRA DOMAIN ASSISTANT
Police Complaint Assistant
Describe what happened in your own language. Accentra will organize it into a clear complaint.
COMPLAINT LANGUAGE
Hindi (हिंदी)
Tap to Speak
Describe what happened naturally. Include details such as your name, location, item, time, or other information if you know them.
"Example: My name is Rahul. Yesterday evening near Chennai Central, someone took my black mobile phone."


Accentra
Home
Translate
Complaint
Saved
History
Settings
ACCENTRA DOMAIN ASSISTANT
Police Complaint Assistant
Describe what happened in your own language. Accentra will organize it into a clear complaint.
YOUR SPEECH (HINDI)
Edit
"मेरा नाम राहुल है कल रात बस स्टैंड पर मेरा मोबाइल फोन चोरी हो गया मेरा फोन कालेले रंग का था और उस समय मेरे पास एक नीली शर्ट पहना हुआ आदमी खड़ा था"
ENGLISH TRANSLATION
"My name is Rahul, my mobile phone was stolen at the bus stand last night. My phone was black and I had a man in a blue shirt standing next to me at the time."





5.2 DISCUSSION

Through the data preprocessing step, there was a balanced set that had 5,075 samples across 24 districts using the technique called the speaker-disjoint partition with zero speaker leakage. With speaker balancing, the model learns the speech accents of the entire region as opposed to just few speakers who are common, whereas speaker disjoint partition ensures that the model evaluates generalization on new speakers.

Since the baseline and adaptive systems are tested on an equal set of test videos, it will help us determine the effects that adapting the system has on WER/CER. Regional adaptation caused a reduction of WER from 30.17% to 28.71%, and CER from 9.03% to 8.67%. These numbers represent a relative improvement of 4.84% in

WER and 3.99% in CER, respectively, thus suggesting that regional adaptation helped the model recognize regionally accented speech without negatively impacting its performance on the multilingual test data.

The efficacy of the output generated from the English complaint process depends on performing all the steps one after the other. If there are any mistakes in the transcription step, the errors could pass into the translation step, and finally, into the extraction process. Therefore, improving the quality of speech recognition brings positive effects for the whole pipeline. Using the transcribed version in English to citizens and providing unsupported fields with blank entries are conscious decisions that lead to errors being visible instead of remaining hidden from sight. Utilization of the GLiNER tool and the distilled version of the IndicTrans2 model keeps the processing approach as CPU-based, avoiding any external generative AI API.

CHAPTER 6
CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

Our work introduces Accentra, a speech accent adaptive and multi-language translator, alongside complaint facilitator for Tamil, Hindi, Telugu, and Malayalam users. For the creation of our region-specific dataset containing 5,075 audio recordings, AI4Bharat IndicVoices is used with six districts for each language with maximum allowed clips per speaker and with the speaker-disjoint split guaranteeing zero speaker leakage. An existing multilingual Indic Conformer ASR is pre-trained to act as a base automatic speech recognition (ASR) model for which we perform fine-tuning with a region-specific dataset. In addition, baseline model and its fine-tunings are tested on an unseen-speaker data through WER & CER metrics.

The modified ASR system has been utilized in combination with IndicTrans2 to translate from Indic languages to English and with GLiNER to detect complaint-based information. Apart from providing an overall speech translation, it converts the citizen's complaint spoken in native languages to complaint details, a concise description, and complaint document format in English without creating any missing details within the complaint. This whole process does not rely on GPUs or any internet-based AI systems because of its dependency on the CPU only.

6.2 FUTURE WORK

- Increase support in the system for more of the 22 official languages covered under IndicConformer and IndicTrans2 systems.
- Increase the amount of adaptation information along with its geographical coverage to cover more districts in every language.
- Make better provision for code-switching, for instance, through modifying models based on code-switched audios.
- Tweak the extraction or translation models with complaint-related texts for better domain vocabulary.
- Create an annotated speech corpus of grievances to measure the performance of extraction in each field individually.

- Incorporating speech synthesis in the citizen's mother tongue will allow the police personnel to ask further queries.
- Create the draft of the complaint based on the format followed by the police station, and ensure two-way communication.
- Consider quantization of models as another way to decrease CPU latencies and use less memory.

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